

Project Deliverable: Working Model & Presentation

1. Project Summary & Problem Statement

- **Project Overview:** Provide a high-level introduction to your project. Clearly state the business problem or research question you are addressing.
- **Dataset Description:** Detail the data used in your project. Describe its source, size, and the key variables (features) it contains.

2. Data cleaning & Pre-processing

- **Data Cleaning:** Explain the methodology for data cleaning. Discuss how you handled missing values, corrected data inconsistencies, and addressed duplicate records.
- **Feature Engineering:** Detail any new variables you created from the raw data. Justify why these new features were necessary for your analysis or model.
- **Data Transformation:** Describe all steps taken to prepare the data for modeling, such as scaling, normalization, or one-hot encoding, and explain the rationale for each.

3. Exploratory Data Analysis (EDA)

- **Key Insights from Descriptive Statistics:** Present a summary of your data's statistical properties. Use measures of central tendency (e.g., mean, median) and dispersion (e.g., standard deviation) to provide a clear picture of the data's distribution and characteristics.
- **Correlations and Relationships:** Discuss the relationships between key variables. Highlight any significant correlations or patterns discovered that are relevant to the problem statement.

4. Machine Learning Model & Evaluation

- **Problem Classification:** Clearly classify the type of machine learning problem you are solving (e.g., classification, regression, clustering).
- **Model Selection:** Justify your choice of machine learning models. Explain which models you experimented with (e.g., Logistic Regression, Random Forest, Gradient Boosting) and why you ultimately selected the best-performing one.
- **Evaluation & Validation:** Explain the criteria used to evaluate your model's performance. Detail the specific metrics (e.g., Accuracy, Precision, Recall, F1-Score for classification; RMSE, MAE for regression) and the validation techniques (e.g., cross-validation, train-test split) you employed to ensure the model's robustness and reliability.

5. Conclusion, Recommendations & Future Work

- **Main Findings:** Summarize the most significant results of your analysis and modeling. Revisit the initial problem and state your final conclusion clearly and concisely.
- **Business Recommendations:** Translate your findings into actionable recommendations for stakeholders. Based on your model's insights, what specific steps should be taken?
- **Limitations & Future Work:** Acknowledge the limitations of your current approach, such as data constraints, model performance, or a specific algorithm's weaknesses. Propose concrete steps for future work to enhance the project, improve the model, or expand the analysis.